



THE UNIVERSITY OF
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SCHOOL OF AEROSPACE, MECHANICAL AND MECHATRONIC
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AERO2460: AEROSPACE DESIGN 1

ASSESSMENT 1B: GROUP DESIGN ASSIGNMENT (PART 2)

Design of an L-Bracket for Aerospace and Automotive Applications

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Abstract

The purpose of this project is to design, develop, and test two 3D-printed L-brackets for both an aerospace and automotive application. These designs were then tested in a UTM and compared to a benchmark design. Both designs outperformed the benchmark.

The aerospace bracket was shaped to carry load mainly through a triangulated, axial path to improve stiffness and capacity. The automotive bracket kept a simpler brace layout to shorten print time while still lifting performance over the benchmark. All brackets were printed to a common baseline and tested under displacement control using a standard cantilever setup. We recorded peak load and tip deflection, and used a cost metric to weigh performance against print time for each application.

Results show both concepts exceeded the benchmark in strength and stiffness. Failures started at the brace/strut roots, which is consistent with stress concentrations. Overall, the aerospace bracket is stronger but takes longer to print, while the automotive bracket is a good choice when faster printing is the priority.

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1 Introduction

Lightweight bracket design is a constrained multi-objective problem: maximise specific strength and stiffness, while minimising mass, manufacturing cost, and print time. In this report, the results of the Trade-Off Study in Assignment 1a are extended by selecting and testing the two most optimally designed PLA L-brackets for an aeronautical and automotive application, respectively. Starting from a benchmark L-bracket that sustains 51 N, candidate designs are required to exhibit failure loads of at least 75 N. The aerospace variant prioritises structural performance with a lower priority on manufacturing costs, whereas the automotive variant balances performance with manufacturing cost (approximated by 3D-printing time). The primary goal of this report is to experimentally validate the analytical calculations performed on the aerospace and automotive candidate brackets via load-displacement testing, computing efficiencies relative to the benchmark, and analysing failure modes.

Overall, the report details the selected bracket features and experimental method, presents load-displacement results and benchmarked efficiencies, and concludes with a discussion of failure modes, requirement compliance, and a discussion of methods for further design optimisation.

2 Design Features

2.1 Aerospace Bracket

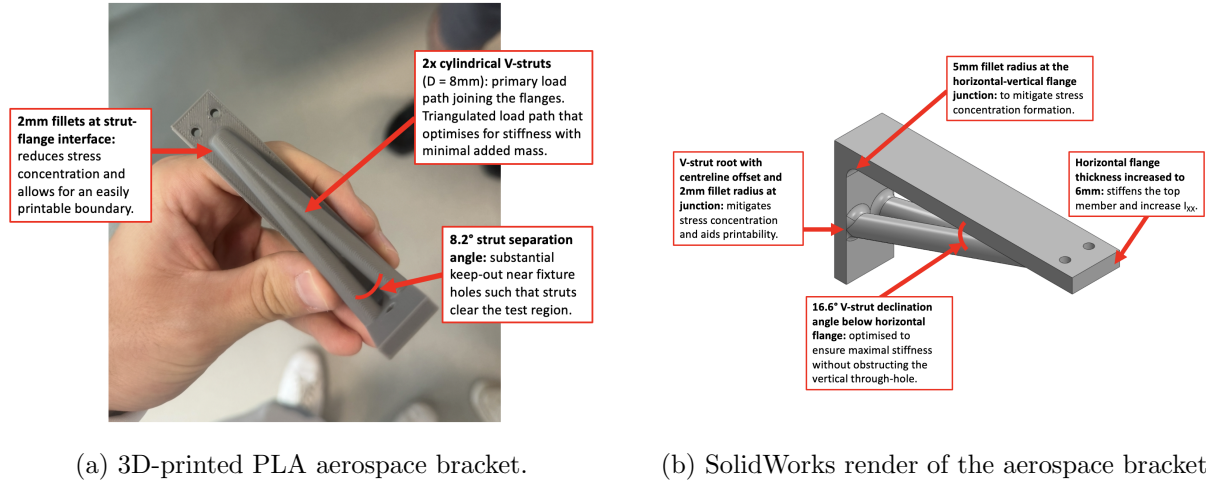


Figure 1: Aerospace bracket geometry and key features. Optimised parameters: strut radius $R = 4.00$ mm, horizontal flange thickness $t_h = 6.00$ mm, strut half-separation $\phi = 4.10^\circ$, inclination $\theta = 16.60^\circ$; junction fillets $R = 2$ mm with a $R = 5$ mm vertical-horizontal corner fillet.

For the aerospace variant of the L-bracket, structural performance was prioritised (maximising specific strength, stiffness and failure load) while adhering to the fixed interface geometry for the experimental setup. The optimised design converts the cantilevered flange into a triangulated load path using twin V-struts and a stiffened top flange, with generous fillets and strict keep-outs to minimise stress concentrations and maintain printability and test compatibility. The following design features were implemented:

- **Twin tapered V-struts (primary load paths):** two cylindrical struts of 8 mm diameter connect the vertical and horizontal flanges, pitched down from the flange at $\theta = 16.6^\circ$

and diverging in-plane by $2\phi \approx 8.2^\circ$. This aligns the members with the principal compression/tension paths while increasing the effective section moment of inertia I_{xx} along the span. The centreline offset beneath the flange grows as $z \tan \theta$ and the in-plane separation as $\Delta x(z) = 2z \tan \phi$, stiffening the bracket towards the tip without obstructing loading holes. The V-struts are positioned to maintain a clear keep-out around the holes, ensuring compatibility with the test rig and reliable load transfer.

- **Stiffened horizontal flange:** the top member is thickened to $t_h = 6$ mm to raise I_{xx} and reduce deflection under tip loading. Under the prescribed 75 N condition, Finite Element Analysis (FEA, Appendix B.1) predicted a maximum deflection of 0.734 mm, confirming the stiffness gain relative to the benchmark.
- **Filleted junctions for durability and printability:** local blends of $R = 2$ mm are applied at strut roots with an enlarged $R = 5$ mm fillet at the vertical–horizontal corner to mitigate stress concentrations and avoid sharp notches in the 3D-printed PLA. At 75 N, peak von Mises stress at the junctions was 13.61 MPa (below the PLA yield strength of 26.08 MPa), indicating elastic behaviour at the minimum permissible design load.

The geometry was selected using a grid-search optimisation that maximised strength- and stiffness-to-mass figures of merit under the 75 N minimum failure-load constraint and the fixed interface geometry. The search spanned strut diameter (D), inclination (θ), half-separation (ϕ), and flange thickness (t_h), with a Pareto-efficient configuration then chosen for manufacture. For printing, a 40% infill was adopted to minimise print time and material usage while preserving shell continuity and the intended load paths.

2.2 Automotive Bracket

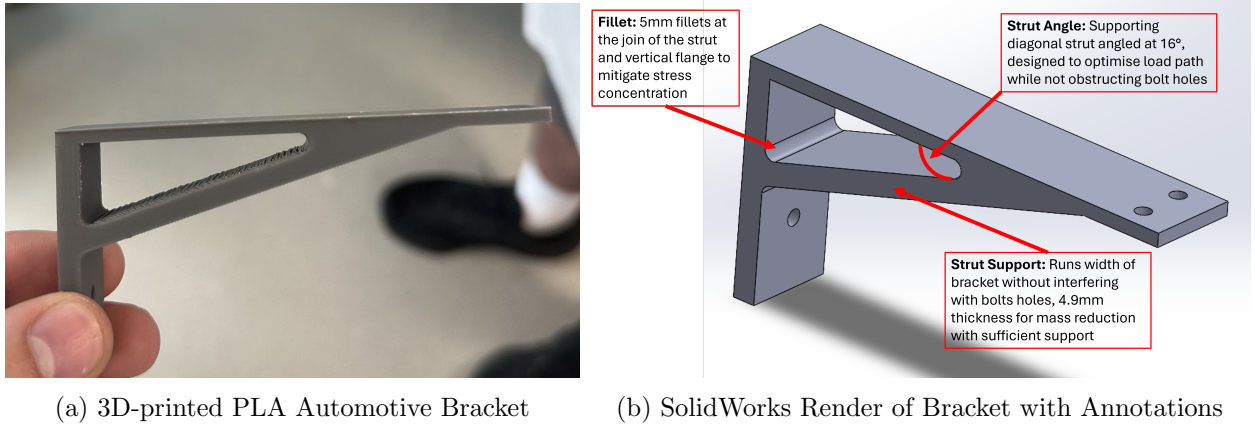


Figure 2: 3D-printed and SolidWorks model for Automotive Bracket

For the automotive bracket variant, priorities were sufficient performance but with low cost, quantified via print time. Similarly to the aerospace bracket, the baseline design was modified to include a diagonal strut support in order to triangulate the load path. This was achieved through the implementation of 4.9mm thick, 20mm wide rectangular support set at a 16° angle to the horizontal flange. Fillets were also employed at several flange intersection points to reduce stress concentration. The vertical and horizontal flanges remained unchanged in an effort to reduce overall weight, print time and complexity. These design changes, shown in figure 2b, are detailed subsequently:

- **Diagonal Strut Support:** The 20mm x 79mm x 4.9mm rectangular support was implemented for load path triangulation. This design ensures sufficient load bearing capacity without drastic increases to print time or mass. Furthermore, it avoids any interference with the bolt locations for testing. It is set at a 16° angle, again for load bearing optimisation and testing configuration requirements.
- **Filletted Flange Intersections:** The intersections of the strut support with both the horizontal and vertical flange are accompanied by a fillet of radius 5mm. This minimises the risk of stress concentrations and avoids sharp edges in the print model.
- **40% Infill Factor:** The 3D-printed model was set with an infill factor of 40%, selected to reduce print time and component mass, thereby minimizing material and the manufacturing cost function. This level of infill preserves shell continuity and provides adequate strength along critical load paths, offering an optimal balance between structural rigidity and fabrication efficiency.

3 Experimental Methodology

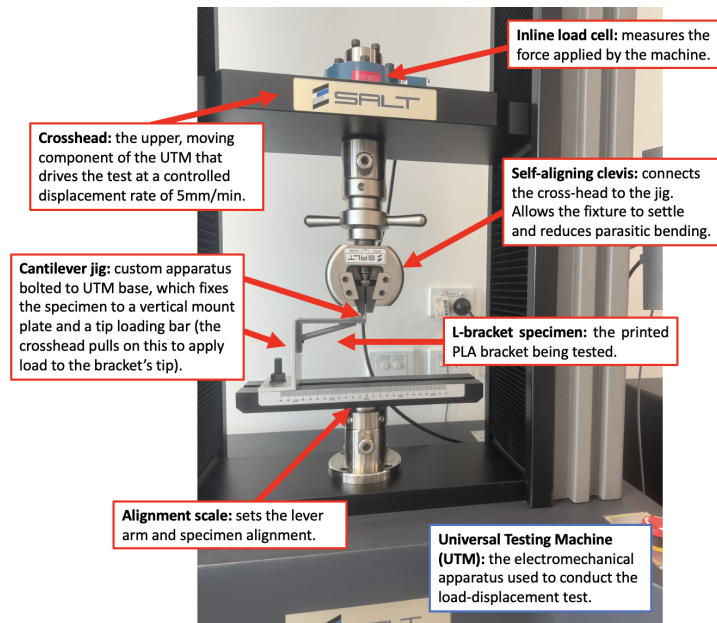


Figure 3: Salt Universal Testing Machine (UTM) apparatus: a displacement-controlled crosshead loads a custom cantilever jig via an inline load cell and self-aligning clevis; the bracket bolts to the vertical plate, the tip engages the loading bar, and the scale sets the lever arm and alignment.

All tests were carried out on an electromechanical universal testing machine (UTM) using the cantilever jig shown in Fig. 3. Each bracket was bolted to the vertical mount plate using the specified hole pattern. The tip flange was connected to the loading bar so that crosshead motion produced a downward load at the tip while the vertical flange acted as the fixed boundary. The machine ran in displacement control at a constant loading rate of 5 mm/min. Load was measured by the inline load cell and displacement was recorded by the controller. A self-aligning clevis allowed the fixture to settle and minimised unintended bending. The loading rate of 5 mm/min provides a very low strain rate for the part and material, so inertia and rate effects are negligible and the

test is treated as quasi-static, consistent with the assumptions used for the analytical and FEA predictions from Assignment 1a.

The following pre-test steps were conducted:

1. Install the L-bracket specimen in the test-jig, verify clearance around the holes and keep-outs, and tighten to the specified torque. Both aerospace and automotive brackets were printed in PLA at 40% infill density.
2. Bring the crosshead into light contact with the loading bar and apply a small seating preload to remove slack.
3. Zero the load channel on the controller and zero the displacement (crosshead position).

The following steps were then taken during the quasi-static load-displacement test:

1. Drive the crosshead at the 5 mm/min loading rate under displacement control until failure of the L-bracket specimen.
2. Record force and displacement continuously at the controller sampling rate with the inline load cell and store this on a computer.
3. After the test, unload, remove the specimen, and document the failure mode and location.

4 Results

Figure 4 demonstrates the relationship between the displacement of the horizontal flange's mounting point against the load applied to both the aeronautical and automotive L-bracket configurations.

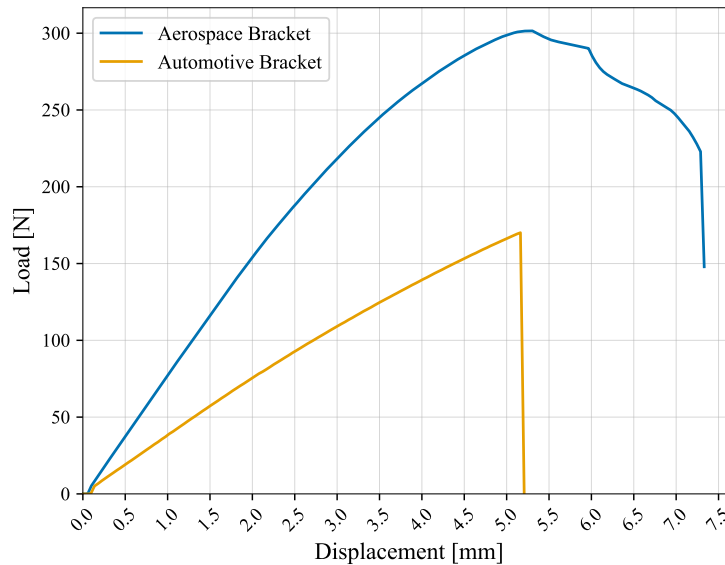


Figure 4: Load vs displacement curve for both test configurations.

4.1 Aerospace Bracket: Comparison to Theoretical Results

The failure load of a specimen is defined as the maximum load point beyond which the structure starts softening. From Figure 4, the failure load of the aerospace bracket was recorded as $P =$

301.5 N, while the deflection at failure was $\delta = 5.26$ mm. For the experimental cost function and efficiency calculations, it is assumed that the benchmark specific strength G , the benchmark 3D printing time T_b , and the fuel efficiency constant F_{aero} are as specified in Table 4 of Appendix A. Further, the 3D print time of the experimental design (T_d) is 4740 s, as determined by the Prusa slicing file. The mass of the experimental design is assumed as $m = 17.71$ g (Appendix B.1), as this accurately accounts for the 40% PLA infill density. The strength cost function (C_1), rigidity cost function (C_3), and the overall efficiency of the experimental aerospace bracket are computed using Equations (1)-(3) and compared to the analytical calculations from Appendix B.1 in Table 1.

$$C_1 = 1.5 \left(\frac{P/m}{G} \right) - \frac{T_d}{F_{\text{aero}} T_b} = 1.5 \left(\frac{301.50/17.71}{51.27/13.45} \right) - \frac{4740}{200 \times 3600} = 6.69 \quad (1)$$

$$C_3 = 1.5 \left(\frac{P/\delta}{S} \right) - \frac{T_d}{F_{\text{aero}} T_b} = 1.5 \left(\frac{301.50/5.23}{51.27/57.76} \right) - \frac{4740}{200 \times 3600} = 97.41 \quad (2)$$

$$\varepsilon_{\text{aero}} = \frac{C_1}{1.5} = \frac{6.69}{1.5} = 4.46 \quad (3)$$

Table 1: Comparison of cost functions and bracket efficiency for the aerospace bracket.

Bracket	P [N]	δ [mm]	C_1	C_3	$\varepsilon_{\text{aero}}$	MOS @ 75 N
Theoretical	270.93	4.50	6.01	101.71	4.01	1.41
Experimental	301.50	5.23	6.69	97.41	4.46	1.68
% Error	11.28	-16.22	11.31	-4.23	11.22	19.15

The aerospace bracket carried about 11% more load than predicted and failed at roughly 16% higher deflection. As a result, C_1 and $\varepsilon_{\text{aero}}$ are higher, C_3 is slightly lower, and the margin of safety at 75 N improves from 1.41 to 1.68. The extra capacity is likely due to conservative material inputs and as-printed details (solid perimeters and fillets) that strengthen the load path. The higher deflection indicates the tested bracket was more compliant than assumed in the analytical model. The analytical model does not explicitly include infill architecture, layer adhesion, small thickness/fillet tolerances, or fixture/joint compliance, all of which reduce effective stiffness. Despite these effects, the analytical predictions accurately captured the overall L-bracket performance parameters to within approximately 5–16% of the experimental values, as exhibited in Table 1.

4.2 Automotive Bracket: Comparison to Theoretical Results

Figure 4 shows the load against displacement curve for the automotive bracket, indicating its failure load at $P = 170.1$ N with a displacement of $\delta = 5.16$ mm. This bracket has lower performance than its aerospace counterpart. Furthermore, its performance is also lower than that predicted by theory, with a roughly 37.9% error observed between predicted and experimental load, as shown by table 2. The experimental deflection was almost 4mm greater than expected. These errors led to similarly high error in predictions of C_2 and C_4 , and a 26.5% lower value for $\varepsilon_{\text{auto}}$, with calculations for these values shown by equations 4, 5 and 6.

$$C_2 = 1.5 \left(\frac{P/m}{G} \right) - \frac{T_d}{F_{\text{auto}} T_b} = 1.5 \left(\frac{170.10/16.23}{51.27/13.45} \right) - \frac{2880}{2 \times 3600} = 4.58 \quad (4)$$

$$C_4 = 1.5 \left(\frac{P/\delta}{S} \right) - \frac{T_d}{F_{\text{auto}} T_b} = 1.5 \left(\frac{170.10/5.16}{51.27/57.76} \right) - \frac{2880}{2 \times 3600} = 55.31 \quad (5)$$

$$\epsilon_{\text{auto}} = \frac{C_2}{1} = \frac{4.58}{1} = 4.58 \quad (6)$$

This can be put down to several factors, particularly 3D-printing parameters that were not properly accounted for. The analytical model does not include variations in strength and stiffness dependent on infill pattern and percentage, layer adhesion, and print orientation. These factors can explain the substantial discrepancy between results. This is further supported by the observed failure mode,

Table 2: Comparison of cost functions and bracket efficiency for the automotive bracket.

Bracket	P [N]	δ [mm]	C_2	C_4	ϵ_{auto}	MOS @ 75 N
Theoretical	273.70	1.35	6.23	342.20	6.23	1.43
Experimental	170.10	5.16	4.58	55.31	4.58	0.51
% Error	-37.85	-282.22	-26.48	-83.84	-26.48	-64.34

with the bracket catastrophically failing due to a fracture below the joint between the diagonal strut and vertical flange. This catastrophic failure is characterised by the immediate reduction to zero of the load at the failure deflection. This is indicative of a bracket intolerant to damage, with failure occurring without warning. This specific failure mode is further evidence of an error in accounting for printing parameters in early analytical processes, with failure occurring due to inter-layer delamination in the print.

Despite these affects, the automotive bracket comfortably exceeded the baseline, with a value for ϵ_{auto} of 4.58. This validates the design method, showing the improvement from baseline performance. Theoretical calculations and other parameters from table 2, such as mass and printing time, obtained from Prusa slicer software, are shown in appendix C.1.

5 Discussion

5.1 Strength and Design Evaluation of Brackets

Table 3 compares each bracket with the benchmark. Both manufactured brackets meet the 75 N requirement under quasi-static loading. The margin of safety is computed as:

$$\text{MOS} = \frac{P_{\text{allow}}}{1.5 \times P_{\text{limit}}} - 1 \quad (7)$$

where the allowable is the measured failure load and the applied load is the specified limit load of 75 N, multiplied by a safety factor of 1.5. The 1.5 factor is required by 14 CFR § 25.303: “*Unless otherwise specified, a factor of safety of 1.5 must be applied to the prescribed limit load.*”[1]

Table 3: Comparison of performance metrics between the manufactured and benchmark designs, with percentage differences between manufactured and benchmark brackets shown for each metric.

Bracket	P [N]	m [g]	T_d [s]	δ [mm]	C_1	C_2	C_3	C_4	ϵ
Aero (Exp.)	301.50	17.71	4740	5.23	6.69	-	97.41	-	4.46
	(+488.06%)	(+31.67%)	(+31.67%)	(-90.95%)	(+347.49%)	-	(+6415.72%)	-	(+347.34%)
Auto (Exp.)	170.10	16.23	2880	5.16	-	4.58	-	55.31	4.58
	(+231.77%)	(+20.67%)	(-20.00%)	(-91.07%)	-	(+358%)	-	(+5431%)	(+358%)
Benchmark	51.27	13.45	3600	57.76	1.495	1.00	1.495	1.00	0.997/1.00

Quantitatively, the aerospace bracket achieves a step change in structural performance relative to the benchmark: peak load increases by roughly +488%, while tip deflection at failure decreases by about 91% (Table 3). These structural performance gains translate into large improvements in

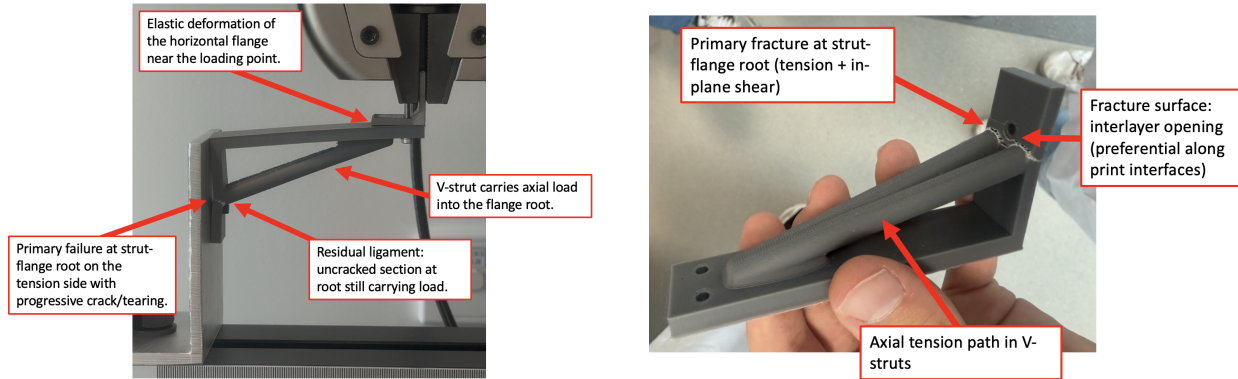
the strength- and stiffness-weighted cost functions (C_1 and C_3) and elevate the efficiency measure $\varepsilon_{\text{aero}}$ to almost 4.5x the benchmark. The increases in mass and print time (both about +32%) are consistent with a performance-first aerospace brief and with standard design practice that prioritises stiffness and strength per unit mass when weight growth and manufacturing time increases are modest [3, 4].

The automotive bracket also significantly exceeds the 75 N requirement with a positive margin (a +231% increase in failure load from the benchmark) and shows a similar reduction in tip deflection (about -91%) relative to the benchmark. However, it reduces print time by 20%. This pattern is reflected in the automotive-weighted cost functions (C_2 , C_4) and the corresponding efficiency $\varepsilon_{\text{auto}}$, indicating that the design meets its brief of equally weighting the structural capacity with manufacturing time.

The test behaviour matches the design intent. The twin V-struts and thicker top flange convert the bracket into a triangulated load path, raise the effective second moment of area, and shift more of the load into axial members rather than pure bending of the flange. This is the response expected from standard performance measures for stiffness and strength per unit mass [3]. Small differences between analysis and test are reasonable for 3D-printed PLA parts, because infill pattern, layer bonding and local joint/fixture flexibility are not fully represented in an idealised beam model and introduce additional compliance and anisotropy [5]. Taken together with Table 3, the increases in strength- and stiffness-weighted metrics (C_1/C_3 for aerospace and C_2/C_4 for automotive) reflect the intended trade-offs: the aerospace design emphasises load-bearing capacity and stiffness at a modest increase in mass and print time, while the automotive design maintains a positive margin with a shorter print time.

5.2 Bracket Failure Modes and Locations

5.2.1 Aerospace Bracket: Observed Failure Modes and Significance



(a) Primary failure at strut-flange root under axial tension in the V-strut. Junction sees net-section tension with peel and in-plane shear over a short run-out; partial load sharing to the opposite strut explains the rounded post-peak response.

(b) Post-test image of the specimen showing fracture at the strut root and interlayer opening along print interfaces. Residual ligament evidence is consistent with gradual capacity loss rather than a brittle drop.

Figure 5: Aerospace bracket: observed tension-driven failure at V-strut roots.

In Figure 4, the load-displacement curve peaks and then decreases slowly, indicating progressive damage rather than sudden failure. The in-jig and post-test failure graphics (Figure 5) confirm that the first damage initiated at the bases of the V-struts where they form a junction with the horizontal flange. Under the applied end load, the V-struts were in axial tension on the underside of

the V-struts/flange and pulled upward at their roots. As this tensile force was transferred into the flange over a short filleted run-out, the junction experienced a combination of net-section tension and peel (opening) plus in-plane shear, typical of a geometry change where an axial member anchors into a plate. In the preliminary design phase, the addition of the 2 mm fillets sought to mitigate the stress concentration. However, Figure 5 indicates that it did not eliminate this entirely.

The fracture surfaces show layer-by-layer opening with limited ductility, consistent with FDM PLA’s anisotropy and interlayer adhesion limits. At the strut–flange root, the tensile strut load is turned into the flange, creating mixed-mode loading: Mode I opening/peel (layers pulled apart) together with Mode II in-plane shear (layers sliding). Under this combination, cracks preferentially follow print interfaces [5, 6]. This explains the rounded post-peak softening in Figure 4: as microcracks formed at one strut root, a residual ligament and the opposite strut continued to carry load, giving a gradual loss of load-carrying capacity rather than instantaneous failure. Mechanically, the V-struts achieved their purpose of increasing stiffness and capacity, but the governing section remained the strut–flange root junction in tension.

From a safety perspective, the behaviour shows some damage tolerance (residual load after first cracking) but the component is arguably not truly fail-safe, because both load paths converge through the same root region. To improve post-damage capability, first spread the tensile load over a longer run-out by increasing the root radius and taper. This is constrained by the loading-hole keep-out on the vertical flange and by the flange width, so very large strut radii are not permissible. Next, add a secondary tie across the root, such as a strap or rib, to create an alternate load path. Finally, raise interlayer tensile and interlaminar shear strength by adjusting print orientation, increasing the number of perimeters, and potentially using higher extrusion temperatures [3, 5].

5.2.2 Automotive Bracket: Observed Failure Modes and Significance

In Figure 4, the load-displacement curve peaks and then drops off sharply, indicating a sudden, critical failure. This critical failure is indicative of a damage-intolerant design, where the system fails catastrophically without prior warning. This is a key aspect of the design of the automotive bracket, which could be improved upon. This failure mode is shown in figure 6. In the figure, we can see a clean horizontal fracture below the joint connecting the strut to the vertical member. This failure is a case of inter-layer delamination, where the applied tensile or shear loads exceeded the adhesion strength between the printed layers. The fracture surface analysis as seen in 6 reveals a clean separation along the horizontal plane. By looking at the part, we can see this is parallel to the print lines along the member. This fracture is characteristic of Mode I, also known as opening mode failure, as the layers of the part have been pulled directly apart. Unlike the progressive failure seen in the aerospace component, which experienced mixed-mode failure, we can see that the automotive bracket’s failure was caused by the print orientation. The part was printed so the principal tensile stress was perpendicular to the printed layers, the weakest possible orientation [2]. This explains the sharp, vertical drop in Figure 4 as once the inter-layer delamination began the part had zero residual load-carrying capacity and failed instantaneously.

From a safety standpoint, this is exemplary damage-intolerant behaviour. The design provides no warning of failure and has no fail-safe characteristics. It failed at the predictable location below the joint due to limitations of its manufacturing process. To improve to a more robust design several changes should be made. First, the part’s print orientation must be changed to be printed on its side. This aligns the stronger extrusion paths with the direction of the stress and thereby moving the load away from the weaker inter-layers [2]. Second, to reduce stress concentration, a larger radius fillet should be added to smooth the load transfer between strut and vertical member.

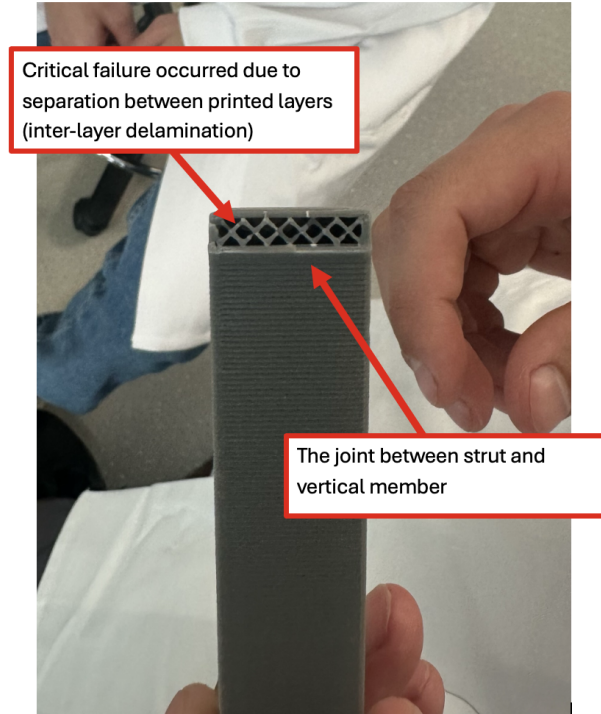


Figure 6: Automotive bracket: observed tension-driven failure at strut root.

5.3 Methods of Further Design Optimisation

Although currently outside the scope of this assignment, further design improvements can be made to both brackets. This section will discuss different techniques that could potentially improve the load-bearing performance of each bracket.

5.3.1 Design for Fused Deposition Modelling

Fused Deposition Modelling (FDM) or Fused Filament Fabrication (FFF) is a common additive manufacturing technology well suited to rapid prototyping, as well as manufacturing complex geometry with significant ease compared to traditional subtractive manufacturing techniques. FDM 3D printing creates parts by extruding a heated thermoplastic in continuous lines across the x-y plane, and building up the geometry of a part layer by layer, laying new beads on top of each previous layer. This process creates highly anisotropic parts, as the thermoplastic has significantly higher strength and stiffness along the length of continuous lines than across layers. Print orientation, amongst a large number of other parameters including; layer height, perimeter count, infill density and pattern, and other printing techniques such as brick layering, can all change the mechanical performance of an FDM 3D printed part significantly.

Both the aerospace and automotive bracket design were observed to fail along layer lines (automotive significantly more than the aerospace) as a result of the principal stress orientation being misaligned with the in-layer higher strength direction of the printed parts. In order to improve performance of each of the brackets, load path analysis should be done to determine the orientation that carries the largest tensile and compressive stresses, and the bracket should be printed in the orientation that aligned the layers with this direction. A limitation of most FDM 3D printers is that only one orientation can make use of this increased strength, and load paths can vary through parts, meaning a choice must be made for the most critical load path. However, experimentation

is currently being done with 3D printers that work in 4 and 5 axis configurations, allowing layer lines to be produced in multiple directions at different areas of a single part.

Additional recommendations for increasing the strength of the bracket through 3D printing techniques include:

- **Increasing perimeter count:** FDM 3D printed parts are printed with multiple lines of extruded filament on the external perimeter of parts, and low density infill inside the perimeters. The perimeters are the primary load bearing part of the structure, and increasing the number of perimeters is directly proportional to increasing the strength of the part.
- **Brick-style print layering:** Brick-style print layering is an experimental technique where instead of having discrete layers across the entire part, layers are extruded at stepped intervals and interwoven - similar to how bricks are laid. This has been shown to increase the strength of parts in the loading direction parallel to layer lines.

As the aerospace and automotive brackets have different requirements and therefore design considerations, additional recommendations are provided specifically for each.

- **Aerospace bracket 3D printing improvements**

- **Different infill patterns:** Changing the infill pattern from *grid* to *gyroid* makes the part have closer to isotropic properties, by printing continuously and weaving infill extrusions in many directions. However, it increases print time, so is less ideal for automotive parts.

- **Automotive bracket 3D printing improvements**

- **Larger nozzle diameter:** By printing with a larger nozzle (compared to the 0.4mm nozzle used in this experiment), larger layer heights can be used, decreasing the number of overall layers and significantly reducing print time.

6 Conclusion

Both brackets met the brief and outperformed the benchmark in strength and stiffness. The aerospace bracket converted the load path to mainly axial through triangulation, which delivered the highest capacity and lowest deflection, but at a longer print time. The automotive bracket used a simpler brace layout that printed faster while still clearing the performance target and beating the benchmark.

Failure consistently initiated at the brace/strut roots with layer separation, indicating stress concentration and highlighting the need for better transitions. Within the limits of FDM (layer adhesion, infill effects, and fixture compliance), the results align with expectations from the design choices.

For applications where strength-per-weight matters, the aerospace bracket is the best option. Where time is the priority, the automotive bracket is the better choice. In future iterations, there would be a larger fillet radius at the roots, and experimentation with print orientation, layer thickness, and perimeter count to improve interlayer strength. These changes target the observed failure modes and should increase the performance of both the aerospace and automotive designs.

References

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Appendix A: Variables and Benchmark Design Parameters

Table 4: Benchmark design and material properties.

Variable	Symbol	Value	Units
Mass	m_b	13.45	g
Load at failure	P_b	51.27	N
Deflection of horizontal section at failure	δ_b	57.76	mm
Time for 3D-printing	T_b	3600	s
Fuel efficiency constant for aerospace	F_{aero}	200	-
Efficiency constant for automotive	F_{auto}	2	-
Elastic modulus of PLA	E	2.1	GPa
Ultimate tensile strength of PLA	σ_{UTS}	36	MPa
Yield strength of PLA	σ_y	26.08 [7]	MPa
Density of PLA	ρ	1265	kg/m ³
Vertical flange width	b_v	20	mm
Horizontal flange thickness	t_h	3	mm
Horizontal flange length	l_h	95	mm

Appendix B: Chosen Bracket Designs and Calculations

The contents of this section are largely adopted from the Appendices in Assessment 1A.

B.1 Aerospace Bracket - Austin Markwick

Figure 7 shows a render of the optimised L-bracket concept; the cross-section and sign convention used in the analysis are given in Figure 8, and a three-view engineering drawing is provided in Figure 9. The design parameters were optimised via a grid search, which sought to maximise the relative load-to-mass ratio and rigidity while minimising printing time and satisfying the 75 N load capacity requirement under room temperature conditions.

The horizontal flange has width b_h and thickness t_h . This design proposes two identical cylindrical struts of radius R which join the vertical and horizontal flanges. The coordinate z measures distance along the horizontal flange from the strut-flange junction toward the fixed vertical flange. For a given load P applied at the tip, the bending moment increases as $M_x(z) = P \cdot z$, so the governing cross-section to optimise for is at $z = z_{\max}$. The struts are pitched by an elevation angle θ measured down from the horizontal flange. The centrelines of the cylindrical struts sit below the horizontal flange by $z \tan \theta$ and are separated in-plane by an angle 2ϕ (assuming a half-angle of ϕ). As shown in Figure 8, this produces a centre-to-centre horizontal separation $\Delta x(z) = 2z \tan \phi$. These angles also set the stress projection along the strut through the directional factor introduced in Equation (12).

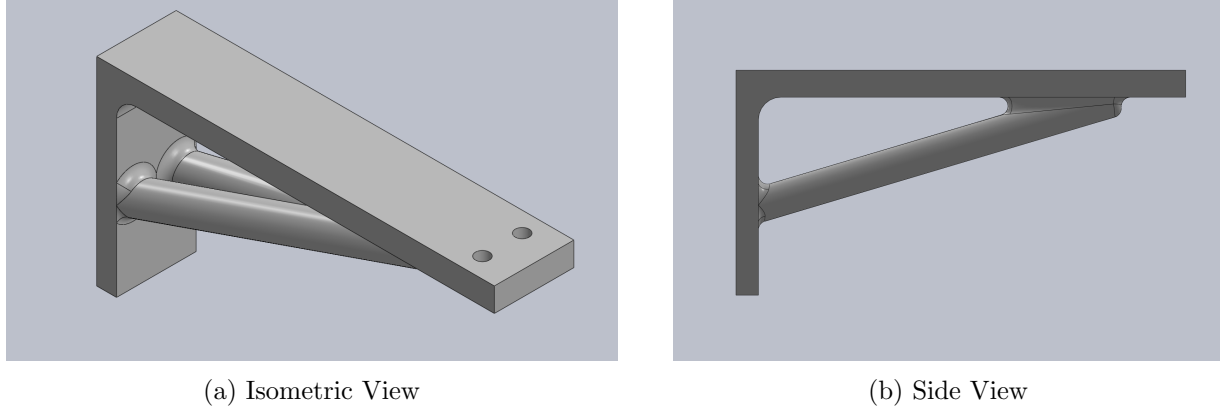


Figure 7: SolidWorks render of Aerospace Bracket Design 1.

Treating the cross section at a general station distance along the horizontal flange, z , as the sum of a rectangle (the flange) and two circles (the cross section of the struts), the y -coordinate of the centroid is given by Equation (8).

$$\bar{y}(z) = \frac{\sum_i y_i A_i}{\sum_i A_i} = \frac{-\frac{1}{2} b_h t_h^2 - 2(t_h + z \tan \theta) \pi R^2}{t_h b_h + 2\pi R^2} \quad (8)$$

Using the centroid position from Equation (8) in the parallel-axis theorem gives the second moment about the moment of inertia about the x -axis is given by Equation (9):

$$I_{xx}(z) = \frac{b_h t_h^3}{12} + b_h t_h \left(-\frac{t_h}{2} - \bar{y} \right)^2 + 2 \left(\frac{\pi R^4}{4} + \pi R^2 (-[t_h + z \tan \theta] - \bar{y})^2 \right) \quad (9)$$

The maximum bending load about the x -axis occurs at the bottom point of the struts, as verified by the Von Mises stress distribution diagrams in Figures 10 under the room temperature conditions,

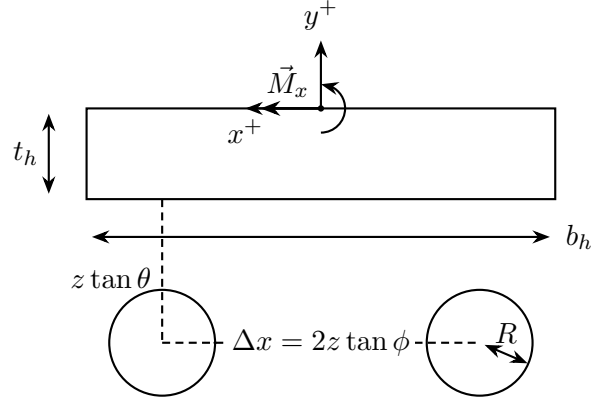


Figure 8: Cross-section schematic of the bracket showing vertical offset $z \tan \theta$, horizontal separation $\Delta x = 2z \tan \phi$, bending moment $\vec{M}_x$ acting in the $+x$ direction, and dimensions b_h , t_h , and R .

respectively. The maximum bending stress in the L-bracket structure is given by Equation (10):

$$\sigma_{z,\max} = \frac{M_{x,\max}}{I_{xx}} y_{\max} = \frac{P_x z}{I_{xx}} \quad (10)$$

For a unit vertical load ($P = 1$) applied to the horizontal flange at the drill holes, so that $M = z$, the bending stress at the bottom fibre of a strut is given by Equation (11):

$$\sigma_z(z)|_{P=1} = \frac{z}{I_{xx}(z)} \left(-[t_h + R + z \tan \theta] - \bar{y}(z) \right) \quad (11)$$

Given that the struts are tapered with respect to the z -axis, the direct stress along the strut is obtained by projecting σ_z onto the strut axis, via Equation (12):

$$\sigma_{\parallel}(z) = \sigma_z(z) \sqrt{\left(\frac{\partial x}{\partial z}\right)^2 + \left(\frac{\partial y}{\partial z}\right)^2 + \left(\frac{\partial z}{\partial z}\right)^2} \quad (12)$$

$$= \sigma_z(z) \sqrt{1 + \tan^2 \theta + \tan^2 \phi} \quad (13)$$

Yield is assumed when $\sigma_{\parallel} = \sigma_y$, which gives the predicted failure load, evaluated at the governing cross-section $z_{\max}$:

$$P_{\text{fail}} = \frac{\sigma_y}{|\sigma_{\parallel}(z_{\max})|_{P=1}} \quad (14)$$

The averaged section moment of inertia was sampled every $\Delta z = 0.5$ mm across the span (161 stations) as

$$\bar{I}_{xx} = \frac{1}{z_{\max}} \sum_k I_{xx}(z_k) \Delta z, \quad (15)$$

and the tip deflection at the effective horizontal flange length L_{eff} of 87 mm followed from the Euler-Bernoulli beam bending theory:

$$\delta \approx \frac{P L_{\text{eff}}^3}{3E \bar{I}_{xx}}. \quad (16)$$

The parameters of the L-bracket were optimised by performing a full-factorial grid search over the domain of plausible designs, which comprised of 152971 possible designs in total:

$$\begin{cases} R \in [1.0, 5.0] \text{ mm} & (\Delta R = 0.1 \text{ mm}), \\ \phi \in [1.0^\circ, 5.0^\circ] & (\Delta \phi = 0.1^\circ), \\ t_h \in [1.0, 10.0] \text{ mm} & (\Delta t_h = 0.1 \text{ mm}). \end{cases} \quad (17)$$

For each candidate design, the pitch angle is set geometrically and capped by the width of the vertical flange:

$$\theta = \min \left\{ \arctan \left(\frac{30 - t_h}{80} \right), 20^\circ \right\}. \quad (18)$$

Any design with $P_{\text{fail}} < 75 \text{ N}$ from Equation (14) is discarded, and selection among the remaining designs is on the basis of the aerospace cost functions C_1 , C_3 , and efficiency $\varepsilon_{\text{aero}}$. Printing time is modelled as proportional to the extruded volume of PLA. Under fixed settings, this scales with mass. As infill is already included in the mass model (fixed at the default setting of 40%), the print time is determined by Equation (19). Because the infill density was fixed at 40%, any reduction in solid geometry or wall thickness directly reduced mass, thereby shortening the print time. The optimiser rejected overly heavy geometries, even if they offered high strength or stiffness, since they incurred a penalty in print time. Conversely, lightweight geometries with struts that were too slender were discarded as they did not achieve the required 75 N load capacity.

$$T_d = T_b \left(\frac{m_{\text{design}}}{m_b} \right). \quad (19)$$

As per Federal Aviation Regulations (FAR) Part 25.303, a factor of safety of 1.5 is applied to the limit loads when calculating the margin of safety (Equation (20)).

$$MOS = \frac{\sigma_y}{\sigma_{z,P=75 \text{ N}} \times 1.5} - 1 \quad (20)$$

The grid search returned an optimum design (at 40% infill) with the parameters outlined in Table 5 and the performance variables outlined in Table 6. As the margin of safety is greater than zero (1.41), the design should effectively withstand the 75 N conditions. The MATLAB code used to obtain these results is listed in Appendix D.1.

Table 5: Optimised geometric parameters (with 40% infill).

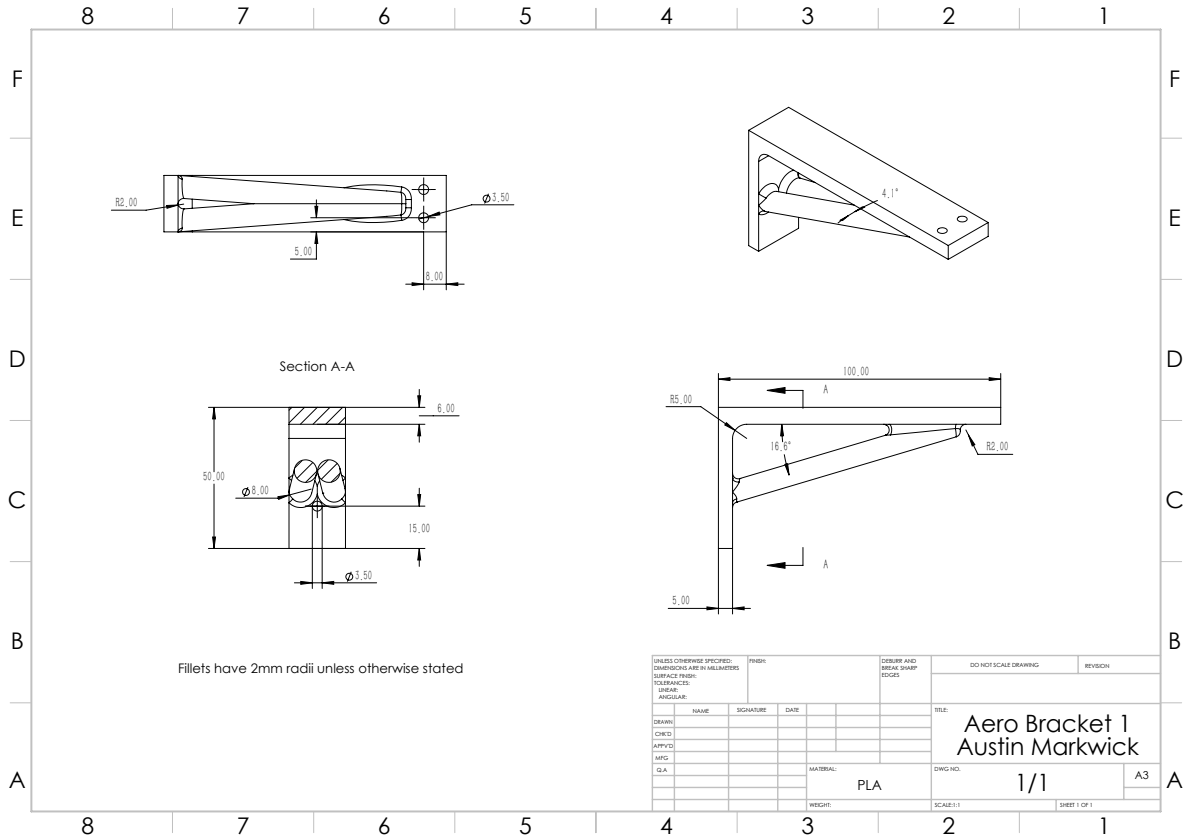
Parameter	Symbol	Value
Strut radius [mm]	R	4.00
Horizontal flange thickness [mm]	t_h	6.00
Strut half-separation [$^\circ$]	ϕ	4.10
Strut inclination [$^\circ$]	θ	16.60

A finite element analysis was conducted to simulate the required 75 N loading condition through the two drill holes on the horizontal flange. As predicted, the maximum stress points occurred at the junctions of the cylindrical V-struts and the horizontal/vertical flanges. The maximum recorded stress was 13.61 MPa (Figure 10), which is significantly lower than the PLA yield strength of 26.08 MPa, indicative of purely elastic deformation in the structure. A 0.734 mm maximum deflection of the structure was produced under the 75 N loading, significantly below the 57.76 mm of the benchmark (Figure 11).

Table 6: Optimised geometry (40% infill) and predicted performance versus benchmark.

Metric	Unit	Proposed Design	Benchmark
Failure load P_{fail}	N	270.93	51.27
Mass m	g	17.71	13.45
Deflection at failure δ	mm	4.50	57.76
Print time T_d	s	4740	3600
Mean I_{xx}	mm ⁴	6.29×10^3	—
Strength cost C_1	—	6.0137	1.495
Rigidity cost C_3	—	101.7109	1.495
Aero efficiency ε_{aero}	—	4.0091	0.9967
MOS @ 75 N	—	1.41	—

Similarly, a von Mises stress and displacement FEA was conducted under the MATLAB-calculated failure load condition of 270.93 N (Table 6) and the results are shown in Figures 12 and 13. On the underside of the V-struts and at the junction points, the tensile stress exceeds the PLA yield stress, indicative of plastic deformation. The ultimate tensile strength of 36 MPa is exceeded, indicating structural failure, with stresses exceeding 49 MPa recorded in and around the bolt holes where the load is applied. The maximum deformation of 2.652 mm at the end of the edge of the horizontal flange (Figure 13) is less than the calculated deflection at failure of 4.50 mm. This suggests that the calculations overestimate the deflection failure, or the deflection measurement in the FEA is invalid given that the ultimate tensile strength has been exceeded in the bracket.



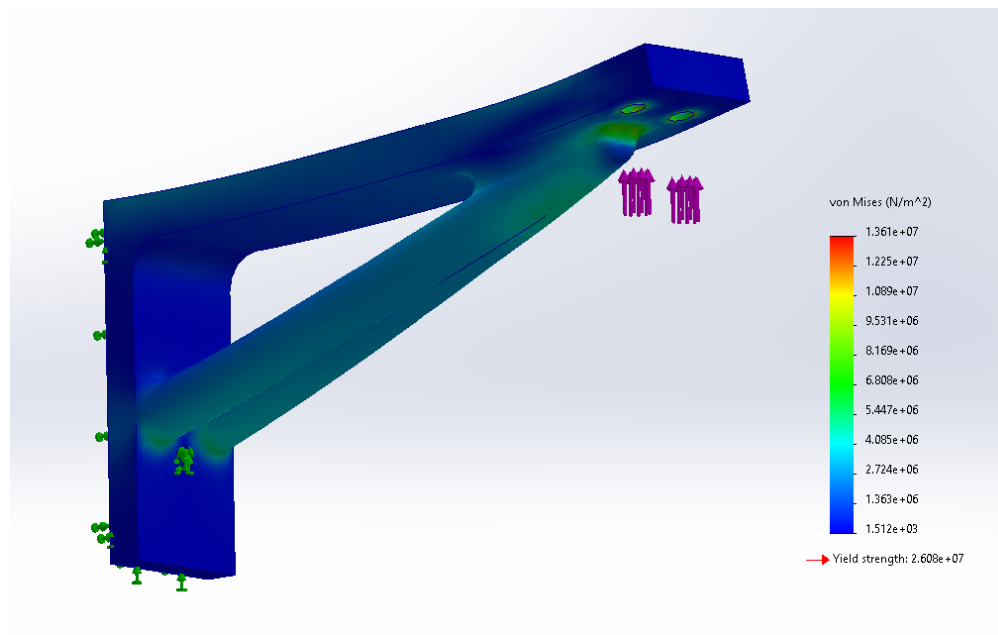


Figure 10: von Mises Stress Analysis with 75 N load applied at the horizontal flange bolt holes.

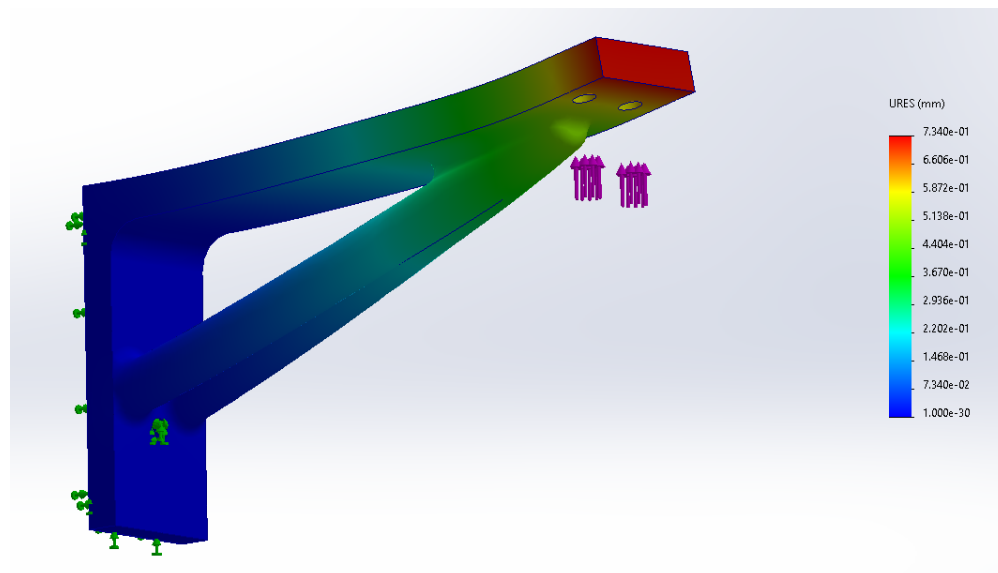


Figure 11: von Mises Displacement Analysis. 75 N load applied at the horizontal flange bolt holes.

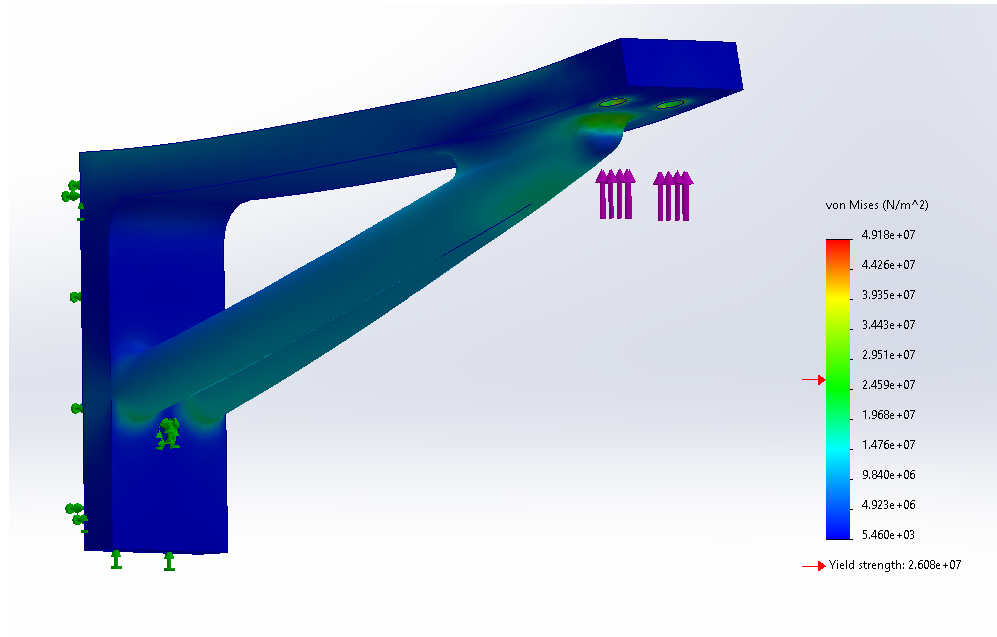


Figure 12: von Mises Stress Analysis with the calculated failure load of 270.93 N load applied.

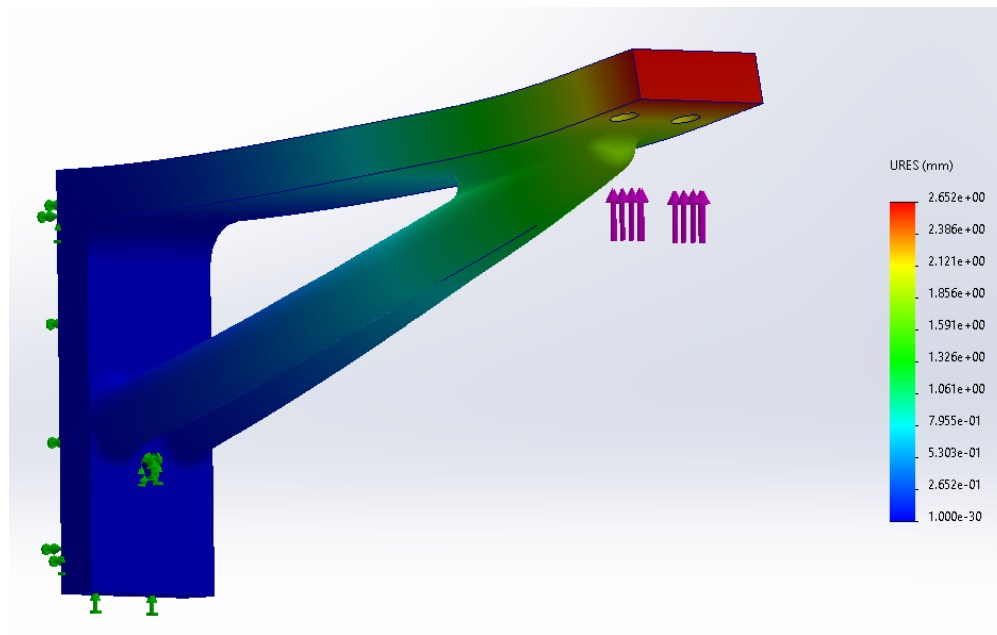


Figure 13: von Mises Displacement Analysis. 270.93 N calculated failure load applied.

Appendix C: Automotive Bracket Designs and Calculations

C.1 Automotive Bracket - Thomas King

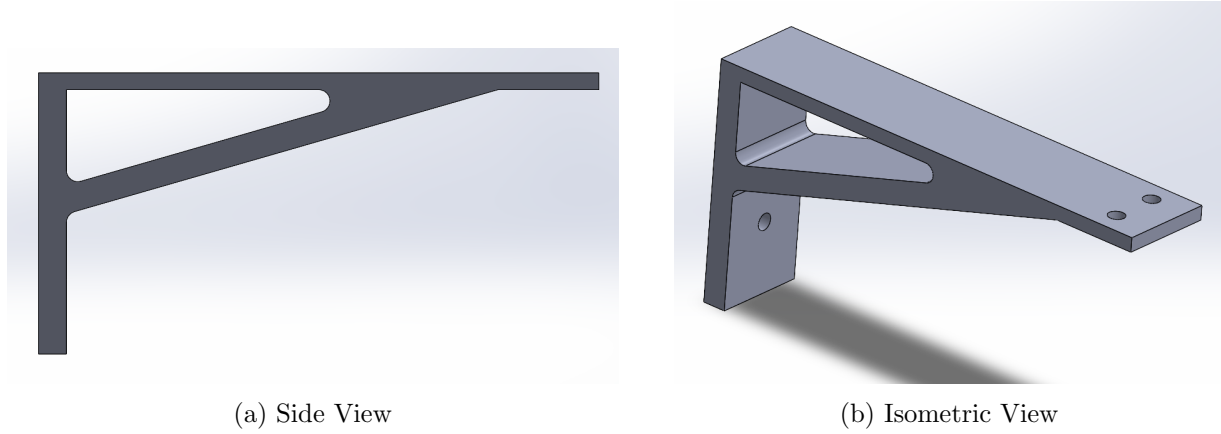


Figure 14: SolidWorks Render of Automotive Bracket

Figure 14 shows the SolidWorks rendering for bracket 4, designed for automotive use. The subsequent section details the calculation of the maximum failure load and other parameters. Figure 15 shows the engineering drawing for this bracket.

All calculations use units of mm, N, and MPa.

Geometry: All members: 20 mm thick

Horizontal flange: 100 mm long $\times$ 3 mm wide $\times$ 20 mm thick

Vertical flange: 50 mm tall $\times$ 5 mm wide $\times$ 20 mm thick

Diagonal brace: 4.9 mm wide $\times$ 20 mm thick $\times$ 78.9 mm long at 16°

Parameters:

$$l_h = 87 \text{ mm} \quad (\text{horizontal distance to load}) \quad (21)$$

$$L_{\text{eff}} = 100 \text{ mm} \quad (\text{effective length}) \quad (22)$$

$$E = 2100 \text{ MPa} \quad (\text{elastic modulus}) \quad (23)$$

$$\sigma_y = 26.08 \text{ MPa} \quad (\text{yield stress}) \quad (24)$$

Cross-sectional areas for root section analysis:

$$A_{\text{hf}} = 20 \times 3 = 60.00 \text{ mm}^2 \quad (\text{horizontal flange}) \quad (25)$$

$$A_{\text{vf}} = 20 \times 5 = 100.00 \text{ mm}^2 \quad (\text{vertical flange}) \quad (26)$$

$$A_{\text{diag}} = 4.9 \times 20 = 98.00 \text{ mm}^2 \quad (\text{diagonal brace}) \quad (27)$$

$$A_{\text{tot}} = 60.00 + 100.00 + 98.00 = 258.00 \text{ mm}^2 \quad (28)$$

Local centroids with datum at top face (positive y downward). For diagonal brace:

$$\text{Vertical projection} = 78.9 \times \sin(16^\circ) = 21.77 \text{ mm} \quad (29)$$

$$\text{Start position} = 3 + (50 - 21.77) = 31.23 \text{ mm} \quad (30)$$

$$\text{End position} = 3.0 \text{ mm} \quad (31)$$

$$y_{\text{diag}} = \frac{31.23 + 3.0}{2} = 17.12 \text{ mm} \quad (32)$$

Component centroids:

$$y_{\text{hf}} = \frac{3}{2} = 1.50 \text{ mm} \quad (33)$$

$$y_{\text{vf}} = 3 + \frac{50}{2} = 28.00 \text{ mm} \quad (34)$$

$$y_{\text{diag}} = 17.12 \text{ mm} \quad (35)$$

Overall centroid:

$$\bar{y} = \frac{A_{\text{hf}} \cdot y_{\text{hf}} + A_{\text{vf}} \cdot y_{\text{vf}} + A_{\text{diag}} \cdot y_{\text{diag}}}{A_{\text{tot}}} \quad (36)$$

$$\bar{y} = \frac{60.00 \times 1.50 + 100.00 \times 28.00 + 98.00 \times 17.12}{258.00} = 17.70 \text{ mm} \quad (37)$$

Second moment of area using parallel-axis theorem.

Horizontal flange contribution:

$$I_{\text{hf}} = \frac{20 \times 3^3}{12} + A_{\text{hf}}(y_{\text{hf}} - \bar{y})^2 = 15793.2 \text{ mm}^4 \quad (38)$$

Vertical flange contribution:

$$I_{\text{vf}} = \frac{5 \times 50^3}{12} + A_{\text{vf}}(y_{\text{vf}} - \bar{y})^2 = 62692.3 \text{ mm}^4 \quad (39)$$

Diagonal brace contribution. Local moments of inertia about diagonal member's centroidal axes:

$$I_{\text{weak}} = \frac{20 \times 4.9^3}{12} = 1960.3 \text{ mm}^4 \quad (40)$$

$$I_{\text{strong}} = \frac{4.9 \times 20^3}{12} = 3266.7 \text{ mm}^4 \quad (41)$$

Transform to global xx -axis using rotation with $\theta = 16^\circ$:

$$I_{\text{diag,local}} = I_{\text{strong}} \sin^2(16^\circ) + I_{\text{weak}} \cos^2(16^\circ) = 2059.6 \text{ mm}^4 \quad (42)$$

Apply parallel axis theorem:

$$I_{\text{diag}} = I_{\text{diag,local}} + A_{\text{diag}}(y_{\text{diag}} - \bar{y})^2 = 2092.5 \text{ mm}^4 \quad (43)$$

Total second moment of area:

$$I_{xx} = 15793.2 + 62692.3 + 2092.5 = 80578.0 \text{ mm}^4 \quad (44)$$

Extreme fiber distances:

$$c_{\text{top}} = \bar{y} = 17.70 \text{ mm} \quad (45)$$

$$c_{\text{bot}} = (3 + 50) - 17.70 = 35.30 \text{ mm} \quad (46)$$

Apply 40% infill correction:

$$\bar{I}_{xx} = 0.4 \times I_{xx} = 32231.2 \text{ mm}^4 \quad (47)$$

Calculate failure load. Unit stress per unit load:

$$\frac{\sigma}{P} = \frac{l_h \times c_{\text{bot}}}{\bar{I}_{xx}} = 0.0953 \text{ MPa/N} \quad (48)$$

Failure load:

$$P_{\text{fail}} = \frac{\sigma_y}{\sigma/P} = 273.7 \text{ N} \quad (49)$$

Tip deflection at failure load using Euler-Bernoulli beam theory:

$$\delta = \frac{P_{\text{fail}} \times L_{\text{eff}}^3}{3 \times E \times \bar{I}_{xx}} = 1.35 \text{ mm} \quad (50)$$

Table 7 shows the maximum load and deflection, amongst other parameters, for the bracket.

Table 7: Bracket geometry (40% infill) and predicted performance versus benchmark.

Metric	Unit	Proposed Design	Benchmark
Failure load P_{fail}	N	273.7	51.27
Mass m	g	16.23	13.45
Deflection at failure δ	mm	1.35	57.76
Print time T_d	s	2880	3600
Mean I_{xx}	mm ⁴	32231.2	—
Strength cost C_2	—	6.23	1.495
Rigidity cost C_4	—	342.2	1.495
Aero efficiency $\varepsilon_{\text{auto}}$	—	6.23	0.9967
MOS @ 75 N	—	1.433	—

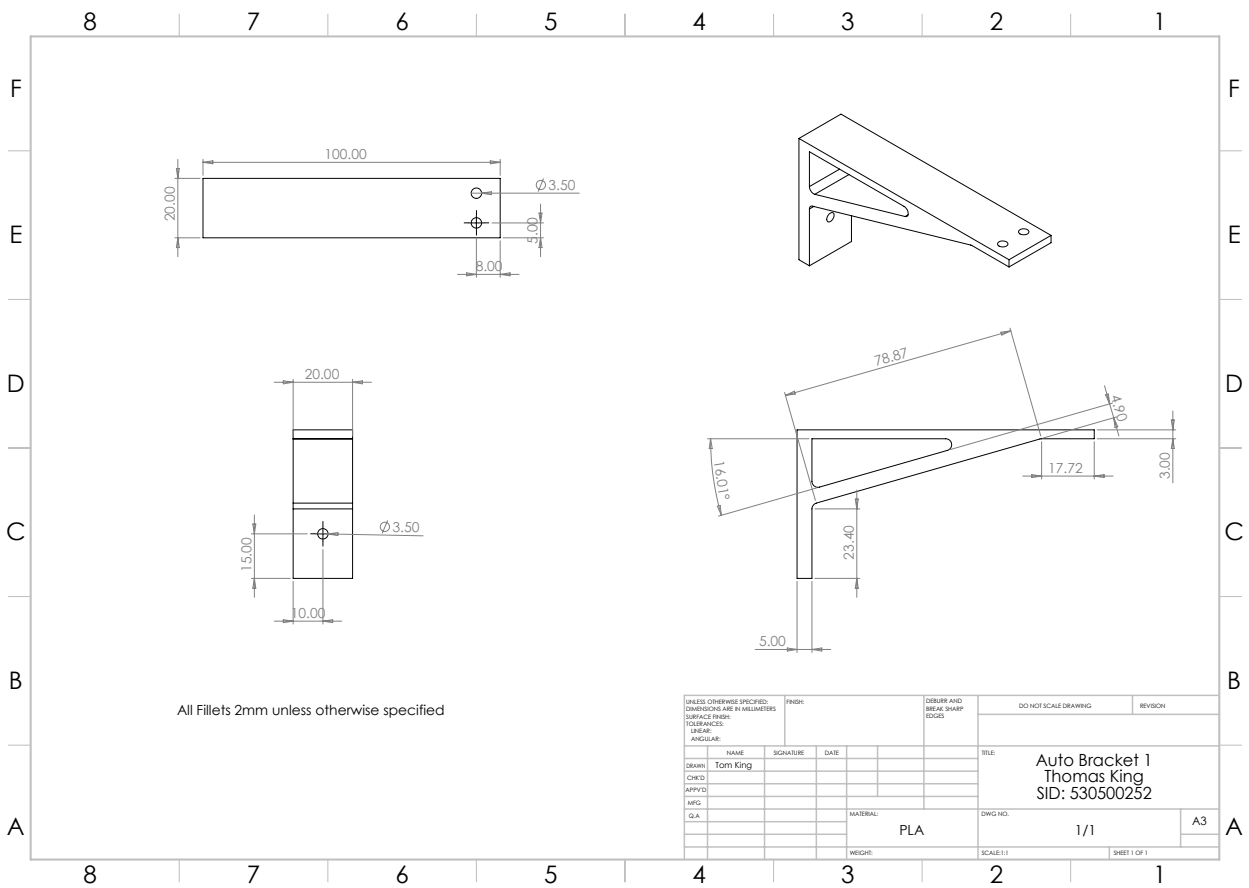


Figure 15: Three-view engineering drawing of Automotive Bracket Design 1.

Appendix D: Design Optimisation Code

D.1 Aerospace Bracket 1 - Austin Markwick

```
clc;
clear;
close all;

% --- Benchmark Data and Material Properties ---

% Material Properties for PLA
E_GPa = 2.1;           % Elastic modulus (GPa)
sigma_uts_MPa = 36;    % Ultimate tensile strength (MPa)
sigma_y_MPa = 26.08;   % Yield strength (MPa), from report [1]
rho = 1265;            % Density (kg/m^3)

E = E_GPa * 1e9;       % Elastic modulus in Pa
sigma_y = sigma_y_MPa * 1e6; % Yield strength in Pa

% --- 3D Printing Parameters ---
infill_density = 0.4;

% Benchmark Bracket Performance
benchmark.P = 51.27;    % Load at failure (N)
benchmark.m_g = 13.45; % Mass (g)
benchmark.m_kg = benchmark.m_g / 1000; % Mass (kg)
benchmark.delta_failure = 57.76 * 1e-3; % Deflection at failure (m)
benchmark.T = 3600;    % Time for 3D printing (s)

% Cost Function Constants
benchmark.F_aero = 200; % Fuel efficiency constant for aerospace

% Calculated Benchmark Metrics
benchmark.G = benchmark.P / benchmark.m_kg; % Specific strength (N/kg)
benchmark.S = benchmark.P / benchmark.delta_failure; % Rigidity (N/m)
benchmark.V = benchmark.m_kg / rho; % Volume (m^3)

% --- Fixed Design Geometry (in SI units) ---
b_flange = 20e-3; % Horizontal flange width (m)
l_h = 95e-3; % Total horizontal flange length (m)
L_eff = 87e-3; % Effective cantilever length for deflection (m)
z_max = 80e-3; % Max z-distance for strut connection (m)
theta_max_deg = 20; % Max allowable strut angle (degrees)
theta_max = deg2rad(theta_max_deg);

% Volume of the fixed vertical member (constant for all designs)
% Dimensions: 50x20x5 mm with one 3.5mm diameter hole
V_vert_flange = (50e-3 * 20e-3 * 5e-3) - (pi * (3.5e-3 / 2)^2 * 5e-3);

% --- Design Parameter Grids for Optimisation Search ---
% These are the variables we can change to find the best design.
strut_radii = (1.0:0.1:5.0) * 1e-3; % Strut radius [m]
phis = deg2rad(1:0.1:5); % Half-separation angle between struts [rad]
```

```

t_h = (1.0:0.1:10.0) * 1e-3;          % Horizontal flange thickness [m]

% Grid for calculating average moment of inertia along the beam
z_grid = 0:0.5e-3:z_max; % Step every 0.5mm as per report methodology

% --- Optimisation Loop ---
% This section iterates through all possible design combinations to find
% the best one.

fprintf('Starting optimisation search...\n');
best = struct('C1', -inf); % Initialize 'best' structure to store the
    optimal design
counter = 0;
total_iterations = length(strut_radii) * length(phis) * length(t_h);

for R = strut_radii
    for phi = phis
        for th = t_h
            % --- Geometry Calculation for current design ---
            % Theta is determined by the connection points, capped at
            % theta_max
            theta_max = atan((30e-3 - th) / z_max);
            % theta = 20;
            theta = min(theta, theta_max);

            % --- Mass Calculation ---
            V_horiz_flange = b_flange * th * l_h;
            L_strut = z_max * sqrt(1 + tan(theta)^2 + tan(phi)^2);
            V_struts = 2 * pi * R^2 * L_strut;
            V_total = V_vert_flange + V_horiz_flange + V_struts;
            % Adjust mass for infill. Note: This is an approximation as
            % the outer
            % shells are typically solid. A slicer gives a more accurate
            % mass.
            m_design = 0.01345 + V_struts * rho * infill_density; % Mass
            % in kg

            % --- Stress Analysis to find Failure Load (P_fail) ---
            % Calculations are based on the cross-section at z_max where
            % moment is highest.
            z = z_max;
            y_bar_zmax = (-0.5*b_flange*th^2 - 2*(th + z*tan(theta))*(pi*R
                ^2)) / (b_flange*th + 2*pi*R^2);
            I_xx_zmax_solid = (b_flange*th^3)/12 + b_flange*th*(-th/2 -
                y_bar_zmax)^2 + ...
                (pi*R^4)/2 + 2*pi*R^2*(-(th + z*tan(theta))
                    - y_bar_zmax)^2;

            % Scale Moment of Inertia by infill density to get effective
            % stiffness
            I_xx_zmax = I_xx_zmax_solid * infill_density;

            % Stress is calculated at the bottom-most fiber of a strut
            y_stress = -(th + R + z*tan(theta))-y_bar_zmax;

            % Stress per unit load (P=1N)

```

```

M_per_P = z; % Moment = P*z, so moment per unit load is just z
sigma_z_per_P = (M_per_P / I_xx_zmax) * y_stress;
sigma_parallel_per_P = sigma_z_per_P * sqrt(1 + tan(theta)^2 +
    tan(phi)^2);

% Predicted failure load based on yield strength
P_fail = sigma_y / abs(sigma_parallel_per_P);

% --- Constraint Check ---
% Skip this design if it doesn't meet the minimum load
% requirement
if P_fail < 75
    continue;
end

% --- Cost Function Calculation ---
% Estimate 3D printing time based on mass ratio, which
% accounts for infill density.
T_d = benchmark.T * (m_design / benchmark.m_kg);

% Aerospace cost function C1 (based on specific strength)
C1 = 1.5 * ((P_fail / m_design) / benchmark.G) - (T_d / (
    benchmark.F_aero * benchmark.T));

% --- Update Best Design ---
if C1 > best.C1
    best.C1 = C1;
    best.R = R * 1000; % store in mm
    best.phi = rad2deg(phi); % store in degrees
    best.th = th * 1000; % store in mm
    best.theta = rad2deg(theta); % store in degrees
    best.P_fail = P_fail;
    best.mass = m_design * 1000; % store in grams
    best.T_d = T_d;

    % --- Deflection Calculation (only for the best candidate
    % to save time) ---
    I_xx_samples = zeros(size(z_grid));
    for i = 1:length(z_grid)
        zi = z_grid(i);
        y_bar = (-0.5*b_flange*th^2 - 2*(th + zi*tan(theta))*(
            pi*R^2)) / (b_flange*th + 2*pi*R^2);
        I_xx_samples(i) = (b_flange*th^3)/12 + b_flange*th*(-
            th/2 - y_bar)^2 + ...
            (pi*R^4)/2 + 2*pi*R^2*(-(th + zi*tan
                (theta)) - y_bar)^2;
    end
    I_xx_avg_solid = mean(I_xx_samples);
    I_xx_avg = I_xx_avg_solid * infill_density; % Scale by
    infill
    delta = (P_fail * L_eff^3) / (3 * E * I_xx_avg);
    best.delta = delta * 1000; % store in mm
    best.I_xx_avg_mm4 = I_xx_avg * 1e12; % store in mm^4

    % --- Additional Cost Function & Efficiency Calculations
    ---

```

```

        P_delta_design = P_fail / delta; % Rigidity of current
            design (N/m). Convert delta back to m
        C3 = 1.5 * (P_delta_design / benchmark.S) - (T_d / (
            benchmark.F_aero * benchmark.T));
        epsilon_aero = C1 / 1.5;

        % --- Margin of Safety Calculation ---
        % MOS = (Allowable Stress / Actual Stress) - 1
        % Actual Stress = Calculated stress at 75N load * Factor
            of Safety (1.5)
        stress_at_75N = abs(sigma_parallel_per_P) * 75; % Stress
            in Pa
        MOS = (sigma_y / (stress_at_75N * 1.5)) - 1;

        best.C3 = C3;
        best.epsilon_aero = epsilon_aero;
        best.MOS = MOS;
    end

    counter = counter + 1;
    if mod(counter, 500) == 0
        fprintf('...processed %d / %d iterations.\n', counter,
            total_iterations);
    end
end
end
end

% --- Display Results ---
fprintf('\n--- Optimisation Complete ---\n');
if isinf(best.C1)
    fprintf('No design found that meets the minimum load requirement of 75
        N.\n');
    fprintf('Consider expanding the parameter grids or checking
        constraints.\n');
else
    fprintf('Optimal Design Parameters Found (at %.0f%% infill):\n',
        infill_density*100);
    fprintf(' - Strut Radius (R):           %.2f mm\n', best.R);
    fprintf(' - Flange Thickness (t_h):       %.2f mm\n', best.th);
    fprintf(' - Strut Half-Angle (phi):             %.2f deg\n', best.phi);
    fprintf(' - Strut Angle (theta):                %.2f deg\n', best.theta);
    fprintf('\nPredicted Performance:\n');
    fprintf(' - Strength Cost (C1):                  %.4f\n', best.C1);
    fprintf(' - Rigidity Cost (C3):                  %.4f\n', best.C3);
    fprintf(' - Aero Efficiency (eps_a):             %.4f\n', best.epsilon_aero);
    fprintf(' - Failure Load (P_fail):              %.2f N\n', best.P_fail);
    fprintf(' - Margin of Safety (at 75N):           %.2f\n', best.MOS);
    fprintf(' - Total Mass:                         %.2f g\n', best.mass);
    fprintf(' - Deflection at Failure:               %.2f mm\n', best.delta);
    fprintf(' - Avg. Moment of Inertia:              %.2e mm^4\n', best.I_xx_avg_mm4
    );
    fprintf(' - Est. Printing Time:                  %.0f s\n', best.T_d);
end

```